

Payne (Pei En) Yeh

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Summary

Software engineer with a background in atmospheric science, applied ML, and data engineering. Experienced in building backend APIs, CI/CD automation pipelines, and embedded IoT systems. Passionate about bridging real-world data infrastructure with AI-powered applications.

Work Experience

Freelance Backend Developer *Apr–Oct 2025*

Python | Flask | PostgreSQL | TDD

- Developed RESTful APIs in Python Flask following Test-Driven Development (TDD), achieving 100% test coverage across all endpoints.
- Designed PostgreSQL database schemas and built query indexes to optimize high-frequency read operations.

Software Development & Cybersecurity Intern *Mar–Jul 2024*

Ernst & Young (EY) Consulting

Python | Django | Selenium | SQL

- Architected an automated bidding document crawler and search system using Django MVC architecture.
- Optimized Python Selenium scraping scripts to process and structure 5,000+ external procurement records.

Selected Projects

Taiwan AI Weather App *Mar–Oct 2025*

Flutter | Python | GitHub Actions | Firebase | Cloudflare R2 | Vertex AI

- Integrated Vertex AI (Gemini) to power an in-app AI weather assistant providing personalized forecasts and radar analysis.
- Implemented Geohash spatial indexing on Firestore, reducing database reads by ~85% across 460+ weather stations.
- Downsampled forecast grids 1×1 km → 5×5 km (~95% smaller payload); local caching cut API calls by ~70%.
- Built GitHub Actions CI/CD pipelines for automated testing, data ingestion, and Telegram alerting.

MQTT-based IoT Smart Classroom *Feb–Aug 2023*

C/C++ | ESP32 | MQTT | Raspberry Pi | YOLOv8

- Developed ESP32 MCU firmware (C/C++) and a Raspberry Pi MQTT broker for real-time IoT sensor control and device automation.
- Integrated YOLOv8 for real-time indoor occupancy counting (95%+ accuracy) to automatically adjust HVAC and lighting.
- Awarded Finalist, National Smart Innovation Cross-disciplinary Competition (2023).

HRV-based Sleep Stage Classification *Feb–Jun 2024*

Python | Scikit-learn | PyTorch | Signal Processing

- Applied Empirical Mode Decomposition (EMD) on PhysioNet clinical dataset (1,985 subjects) to extract time-frequency HRV features.
- Trained Random Forest, Decision Tree, and SVM models for 5-class sleep stage prediction; achieved 82% accuracy on held-out test set.

Education

National Taiwan University

MS in Atmospheric Sciences

Starting Sep 2026

National Central University

BS in Atmospheric Sciences

AI Application Credit Program

Sep 2020 – Jun 2024

Outstanding Student Scholarship (2024)

Skills

- Languages:** Python, C/C++, SQL, JavaScript/TypeScript, Dart
- Backend:** Flask, Django, PostgreSQL, RESTful APIs
- DevOps:** GitHub Actions, Docker, Linux, Firebase, Cloudflare R2
- AI/ML:** PyTorch, Scikit-learn, Vertex AI (Gemini)

Certifications

- Microsoft AZ-900 – Azure Fundamentals
- Microsoft AI-900 – Azure AI Fundamentals

Awards & Competitions

- 1st Place, NCU Sustainable Innovation Proposal Competition (2023)
- 2nd Place, Weather Hackathon (2023)
- Finalist, National Smart Innovation Cross-disciplinary Competition (2023)

Languages

- English: TOEIC 805